# Multi-Modal Demonstrations for Dexterous Manipulation: Combining Teleoperation and Kinesthetic Teaching

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***Abstract*—Collecting demonstrations for long-horizon, contact-rich manipulation remains a major bottleneck in robot learning. Teleoperation scales to long-horizon tasks but is imprecise for fine, contact-rich interaction, whereas kinesthetic teaching excels at contact-rich manipulation but is physically exhausting and, critically, records the demonstrator's hands in every frame. This paper investigates how teleoperation and kinesthetic teaching can be combined for training policies for dexterous manipulation. We propose a demonstration framework that uses teleoperation data for broad task execution and kinesthetic teaching for difficult fine-control phases. We first train a diffusion policy on teleoperated demonstrations, then refine it through real-time kinesthetic alignment. Because the human hand appears during correction but not at deployment, we supervise the visual encoder with object masks so it attends to the manipulated object rather than the demonstrator's hand. We were able to visualize the visual encoder that attends more to the manipulated object. This work suggests that combining heterogeneous demonstration types allows a single policy to handle both long-horizon execution and fine-grained contact-rich phases of dexterous manipulation.**

## I. Introduction

Collecting demonstrations with one modality limits the data a robot can acquire. Each way of demonstrating a skill captures some aspects of a task well and others poorly, so for dexterous, contact-rich manipulation no single modality is sufficient on its own. The two most commonly used modalities are complementary. **Teleoperation** captures the long-horizon structure of a task but is imprecise during contact-rich phases. **Kinesthetic teaching**, in which the demonstrator physically molds the robot's fingers, excels precisely at contact-rich phases, but is physically exhausting and, more fundamentally, records the demonstrator's hands directly in the camera observation.

We combine the two modalities so that each modality can cover different aspects of the task. Teleoperation demonstrations cover the long-horizon task, while kinesthetic corrections supply the precise, contact-rich interactions teleoperation misses. Together, the two modalities let us collect demonstration data that neither captures on its own.

The main obstacle to using this combined data is the visual input. An image-based policy trained on kinesthetic frames sees the demonstrator's hands in every frame, whereas at deployment those hands are never present. This visual domain gap between training and deployment observations causes policies trained naively on kinesthetic data to fail, even when the underlying motion supervision is correct. To close it, we train an image-based diffusion policy with an auxiliary segmentation objective that steers the visual encoder toward the manipulated object rather than the hands. The segmentation targets are produced by SAM-3 and used only during training, since it is inconsistent during deployment.

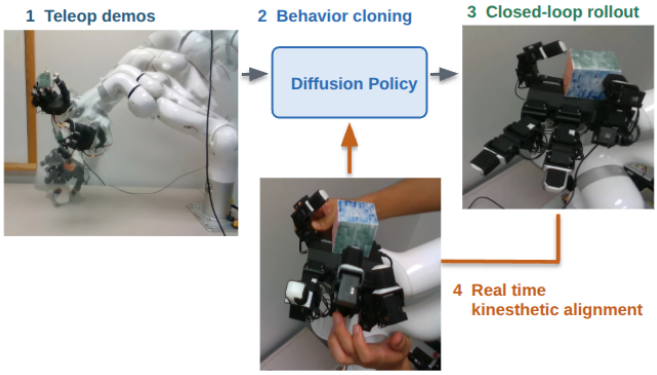



Fig. 1: Overview of our multi-modal demonstration pipeline. We first collect **teleoperation** demonstrations for broad task coverage and train an image-based diffusion policy by behavior cloning. During closed-loop rollout, wherever contact-rich execution fails, we mold the robot's gravity-compensated hand in real time via **kinesthetic teaching** to produce corrective demonstrations. These corrections are aggregated and used to retrain the policy in a DAgger-style loop.

In summary, this paper makes the following contributions:

- We introduce a **multi-modal demonstration pipeline** that combines teleoperation for task coverage with kinesthetic teaching for contact-rich correction. This lets a dexterous manipulator collect demonstration data that no single modality captures.
- To make this kinesthetic data usable by an image-based policy, we identify the **hand-induced visual domain gap** and close it with an **object-focused segmentation objective** that adds consistency.
- We evaluate the approach on **in-hand rotation**, where hand-invariant observations reach an 80% success rate. A Grad-CAM analysis shows this success relies largely on proprioception rather than attentioning to the task relevant regions.

## II. Related Work

### A. Demonstration Collection for Manipulation

Imitation learning is the dominant paradigm for teaching manipulation skills, and its performance is bounded by how demonstrations are collected [1]. **Teleoperation** is the most scalable source of demonstrations, using interfaces such as space mice, VR controllers, and vision-based hand retargeting

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to command the robot [2], [3], [4]. It captures the long-horizon structure of a task across many embodiments, but such interfaces struggle to command the fine, high-bandwidth contact forces that dexterous, contact-rich manipulation requires. Improved retargeting can push this boundary; ResPilot, for instance, learns a residual correction that expands the robot hand's reachable workspace and enables teleoperated in-hand finger gaiting such as cube rotation [5]. Such methods, however, remain tied to a retargeting map from the operator's hand to the robot's joints, which constrains the attainable robot configurations to those the human hand can produce. **Kinesthetic teaching**, in which the demonstrator physically guides the robot, instead yields accurate contact-rich trajectories without an intermediary interface [6], at the cost of physical fatigue and, for image-based policies, occlusion of the scene by the demonstrator's hands. Rather than commit to one modality, we combine both, using teleoperation for task coverage and kinesthetic teaching for the contact-rich data teleoperation misses.

### B. Interactive and Corrective Imitation

Policies trained purely from offline demonstrations accumulate error under distribution shift, motivating interactive imitation that gathers corrective data on the policy's own state distribution. DAgger [7] and its human-gated variants [8] query an expert where the policy fails and aggregate these corrections for retraining. We adopt this interactive philosophy for *contact-rich* correction, molding the robot's fingers kinesthetically in real time during rollout and aggregating the corrections in a DAgger-style loop.

### C. Diffusion Policies for Manipulation

Diffusion policies model the multi-modal distribution of action sequences and have become a strong choice for visuomotor imitation [9], [10]. We adopt an image-conditioned diffusion policy with a conditional 1-D U-Net denoiser producing action chunks [9]. Unlike policies that consume raw observations directly, we augment the visual encoder with an auxiliary object-segmentation objective that controls what the encoder attends to.

### D. Bridging Visual Domain Gaps

A policy transfers reliably only when its observations are invariant to task-irrelevant appearance such as backgrounds, embodiment, or, in our case, the demonstrator's hands. Generative visual augmentation [11] and object-centric representations [12] improve this invariance. Two families of methods remove the offending pixels more directly. The first masks the input, applying a segmentation mask to the observation so that only task-relevant regions reach the policy at both training and deployment [13], [14]. This requires a clean mask at every control step, yet a segmentation model cannot produce one consistently in the loop, and a missed or flickering mask corrupts the very observation the policy depends on. The second inpaints the human out of the frame, as in KineDex [15]. When the demonstrator's hands cover a large fraction of the image, as they do throughout kinesthetic teaching, the inpainter must hallucinate the occluded region and leaves smudged, artifact-ridden frames that no longer match the clean image seen at deployment. We instead use segmentation only as a training signal, steering a ResNet encoder toward the manipulated object with an auxiliary segmentation head supervised by SAM masks [16]. At deployment the raw image is fed directly to the policy with no mask and no inpainting, so no test-time segmentation is required. We compare against an inpainting baseline and analyze encoder attention with Grad-CAM [17].

## III. Method

Our system has four stages, illustrated in Fig. 1: (1) teleoperation data collection, (2) behavior cloning of a diffusion policy, (3) closed-loop rollout, and (4) real-time kinesthetic alignment that produces corrective demonstrations for retraining.

### A. Multi-Modal Demonstration Collection

**Teleoperation demonstrations.** A SpaceMouse drives the 6-DOF arm end-effector while a MediaPipe hand tracker retargets the demonstrator's finger poses onto the multi-fingered LEAP hand. We record RGB observations and full robot proprioception at 30 Hz, collecting 100 demonstrations that cover the long-horizon structure of the task.

**Kinesthetic correction.** Teleoperation alone is imprecise during contact. During closed-loop rollout we mold the robot's fingers by hand in real time. This requires a careful balance. The hand motors must compensate the gravity, so the fingers hold their commanded pose, yet remain soft enough to be easily molded by the demonstrator. The corrections are aggregated with the original demonstrations, and the policy is retrained with the combined dataset. This produces a DAgger-style self-improving loop in which teleoperation teaches the task and kinesthetic corrections repair the contact-rich failures.

### B. Image-Based Diffusion Policy

We learn a visuomotor policy with an image-conditioned diffusion model. A ResNet-18 encodes the RGB observation, and an attention-pooling layer pools its features, which are concatenated with end-effector and hand proprioception to condition a conditional 1-D U-Net that denoises an action chunk. The predicted action chunk consists of the end-effector pose and the hand pose, executed on the real robot. Formally, given observation $\mathbf{o}$ (image features $\phi(\mathbf{I})$ and proprioception $\mathbf{q}$), the U-Net $\epsilon_\theta$ is trained with the standard denoising objective

$$\mathcal{L}_{\mathrm{dp}} = \mathbb{E}_{\mathbf{a}_0,\epsilon,t}\left\|\epsilon - \epsilon_\theta(\mathbf{a}_t, t \mid \phi(\mathbf{I}), \mathbf{q})\right\|_2^2, \tag{1}$$

where $\mathbf{a}_t$ is the action chunk noised at diffusion step $t$.

### C. Closing the Hand-Induced Domain Gap

The key difficulty is that kinesthetic training frames contain the demonstrator's hands, whereas rollout frames never do. We steer the visual encoder to be invariant to this difference through an auxiliary segmentation head. During training, a lightweight segmentation head attached to the encoder predicts

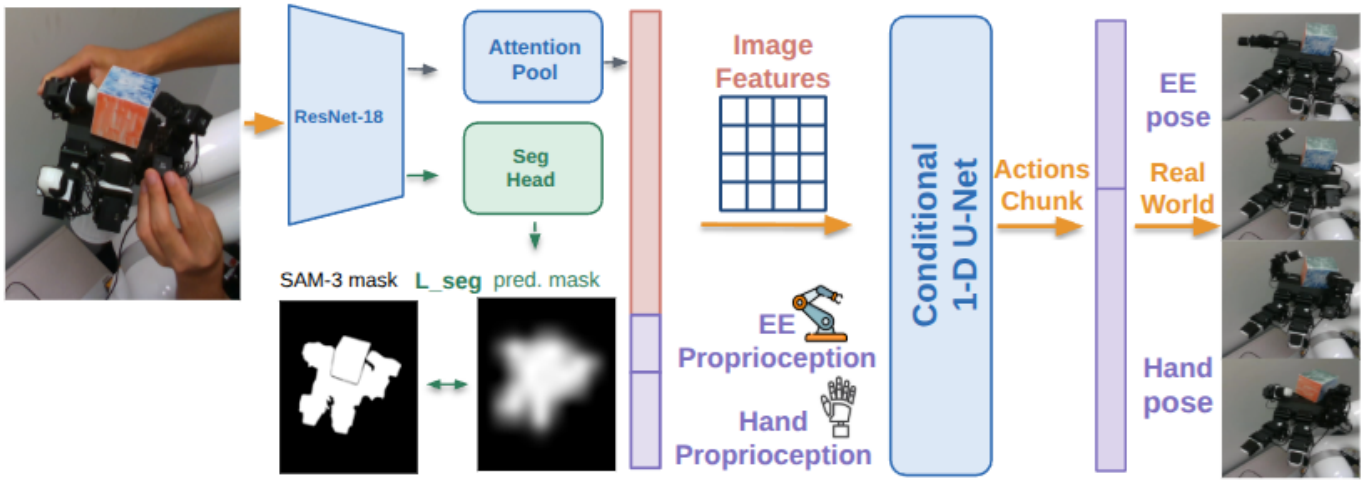



Fig. 2: Overview of the multi-modal demonstration pipeline. Teleoperation demonstrations train an image-based diffusion policy (behavior cloning); the policy is rolled out closed-loop from raw RGB and proprioception; where contact-rich execution fails, the gravity-compensated hand is molded kinesthetically in real time, and these corrections are aggregated and used to retrain the policy in a DAgger-style loop.

the manipulated object's mask, supervised against a SAM-3 object mask $\mathbf{m}^\star$ with a per-pixel loss

$$\mathcal{L}_{\text{seg}} = \text{BCE}\big(\hat{\mathbf{m}},\, \mathbf{m}^\star\big), \qquad \mathcal{L} = \mathcal{L}_{\text{dp}} + \lambda\, \mathcal{L}_{\text{seg}}. \tag{2}$$

By forcing the shared features to be sufficient to localize the object, the objective pulls encoder attention toward the object and away from the hands. Crucially, the segmentation head is used only as a training-time regularizer. At deployment we input the raw RGB image and leave the segmentation head unused, so the policy incurs no additional runtime cost and never requires a mask at test time. Because no mask is generated during inference, the policy is also unaffected by the inconsistent mask quality that degrades methods relying on segmentation at test time.

We contrast this with an alternative that removes the hand from the observation directly by *inpainting* the demonstrator out of each kinesthetic frame before training, producing hand-free images. Both approaches aim at the same goal of hand-invariant observations, and we compare them empirically.

## IV. Experiments

We evaluate on an **in-hand rotation** task in which the robot must reach out, turn its wrist, and rotate a grasped object in-hand. This task couples a long-horizon reaching and reorienting phase with a contact-rich in-hand rotation phase, so a single policy must handle both aspects that our pipeline targets. We ask two questions:

1) Does combining teleoperation with kinesthetic corrections let one policy complete both the long-horizon and the contact-rich phases of the task?
2) Kinesthetic correction records the demonstrator's hands, so can hand-invariant observations recover reliable manipulation despite the resulting visual domain gap?

### A. Setup

The platform is a 6-DOF UFACTORY xArm 850 equipped with a multi-fingered LEAP hand. This experiment isolates the second question, the hand-induced visual domain gap, so we hold the demonstration source fixed and train every policy on kinesthetic demonstrations of in-hand rotation, the modality in which the demonstrator's hands appear in the observation. The full pipeline that also folds in teleoperation demonstrations is still under development, so the comparison here varies only how the training observation treats the demonstrator's hands. Each policy is evaluated over 10 real-world trials with varied object poses, and we report the number of successful rotations across the following three treatments.

- **Current Method**: raw kinesthetic frames, no hand-invariance treatment.
- **Inpainted Image**: the demonstrator is removed from each frame via video inpainting before training.
- **Masked Image**: a segmentation objective explicitly forces the encoder to focus on the object.

### B. Results

On in-hand rotation, both hand-invariant treatments, inpainting and object-focused masking, reach **80%** success over 10 real-world trials, while the raw-frame policy is still under evaluation.

**Attention analysis.** Figure 3 visualizes, via Grad-CAM, where each policy's visual encoder attends. For a target scalar $S$ derived from the policy output, we weight the encoder's final convolutional feature maps $A^k$ by their gradients and combine them into a saliency map,

$$\alpha_k = \frac{1}{Z}\sum_{i,j}\frac{\partial S}{\partial A^k_{ij}}, \qquad M = \text{ReLU}\Big(\sum_k \alpha_k\, A^k\Big), \tag{3}$$

where $Z$ is the number of spatial locations in the feature map and $M$ is upsampled to the input resolution. We take

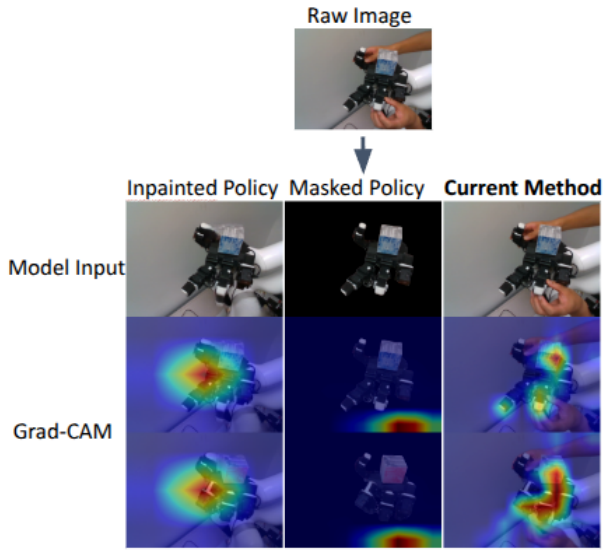



Fig. 3: Grad-CAM attention for the three policies. The raw-frame policy attends to both the cube and the robot hand, entangling its features with the hand seen only during training. The masked and inpainted policies produce nearly identical maps across frames, indicating that they rely mainly on proprioception rather than localizing the cube.

$S = \|\hat{\mathbf{a}}\|_2$, the norm of the predicted action chunk, so that $M$ highlights the pixels that most influence the commanded action.

The three policies attend very differently. The raw-frame policy spreads its attention over both the cube and the robot hand, so its features stay entangled with the appearance of the hand that is present during training but absent at deployment. The masked and inpainted policies instead produce almost the *same* attention map from one frame to the next. Rather than localizing the cube in the image, they appear to rely largely on proprioception to drive the action, with the visual input contributing little.

We probed this by feeding an all-black image to the masked and inpainted policies at deployment, and both failed. The cause is distribution shift rather than genuine image reliance, since an all-black frame lies far outside the training distribution, so this failure does not contradict the finding that these policies depend mostly on proprioception. Taken together, the 80% success rate overstates these policies' visual competence, since they solve the task largely without using the image. We therefore read the current results as a diagnostic of the visual pathway rather than a demonstration of reliable hand-invariant vision.

## V. Conclusion and Future Work

We presented a multi-modal demonstration pipeline for dexterous manipulation that combines teleoperation for long-horizon task coverage with kinesthetic teaching for contact-rich correction. We identified the hand-induced visual domain gap as the reason image policies trained on kinesthetic demonstrations fail, and showed that making observations hand-invariant, either by inpainting the demonstrator out or by an object-focused segmentation objective that adds no deployment cost, reaches an 80% success rate on in-hand rotation. A Grad-CAM analysis, however, shows that these policies lean largely on proprioception rather than the image, so the visual pathway remains to be strengthened. Future work proceeds along three directions. First, we will train the encoder to output the *same* features for images with and without hands, directly enforcing invariance rather than approximating it through an auxiliary objective. Second, we will close the self-improving loop of teleoperation demonstrations, rollout, kinesthetic correction, and retraining into a fully automated DAgger-style pipeline. Third, we will compare the combined multi-modal policy against a teleoperation-only policy to quantify the benefit of kinesthetic correction on contact-rich success.

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